Improved NP-Hardness of Approximation for Max-3-Cut and for Max-k-Cut with Small k

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Abstract

We prove that Max-3-Cut is NP-hard to approximate within a factor of $27/29 + \varepsilon \approx 0.931$. This improves the factor $16/17 \approx 0.941$ of Austrin, O'Donnell, Tan and Wright, the best NP-hardness result for Max-3-Cut since 2012. (Their result, unlike ours, also holds for 3-colourable graphs.) We also improve the NP-hardness of Max-k-Cut for every k from 4 to 20; for instance, we obtain $45/47 \approx 0.957$ for Max-4-Cut and $33/34 \approx 0.971$ for Max-5-Cut, where the previous factors were 0.987 and 120/121.

Our reductions are gadget reductions from Håstad's linear equations over $\mathbb{Z}_3$, with gadgets that have at most six auxiliary vertices. The main new idea is a *negation trick*: since the negation of an assignment is again an assignment, equations with right-hand sides t and $-t$ can be handled together, and this makes the resulting predicates symmetric under all permutations of the colours. We complement it with a general soundness criterion for such reductions: in effect, the best strategy of the adversary is to use a single colour pattern everywhere. This criterion is the strongest analysis that uses Håstad's theorem as a black box, and, using LP duality, we show that under it no gadget reduction from these equations can beat 27/29 for Max-3-Cut.

Finally, we observe that a classical reduction from Max-3-Cut to Max-k-Cut, applied more carefully, already gives $82/85 \approx 0.965$ for Max-4-Cut on 4-colourable graphs, improving the previously stated 85/86.

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1 Introduction

In Max-k-Cut, we are given a graph with non-negative edge weights and must colour its vertices with k colours so as to maximise the total weight of edges whose endpoints receive different colours. For $k = 2$ this is Max-Cut. For $k \geq 3$ the best approximation algorithms round the semidefinite relaxation of Frieze and Jerrum [FJ97]; they achieve a ratio of about 0.836 for $k = 3$ and 0.857 for $k = 4$ [GW04, dKPW04]. For $k = 3$ this is optimal under the Unique Games Conjecture [IM12, Hei26].

Much less is known about NP-hardness. Kann, Khanna, Lagergren and Panconesi [KKLP97] proved hardness of the form $1 - \Theta(1/k)$, and Guruswami and Sinop [GS13] proved bounds of the same form even for k-colourable graphs. The current record for $k = 3$ was set by Austrin, O'Donnell, Tan and Wright [AOTW14]: using a PCP designed for the purpose, they showed that Max-3-Cut is NP-hard to approximate within $16/17 + \varepsilon$, even on 3-colourable graphs. Combined with a reduction of Guruswami and Sinop, the same result gives bounds for every $k \geq 4$, such as $85/86$ for Max-4-Cut [AOTW14]; on k-colourable graphs, and on general graphs for $k \geq 5$, these are the best previously stated bounds. Most recently, Nagda, Raghavan and Thakurta [NRT25] used the AlphaEvolve system [N$^+$25] to search for gadget reductions from Håstad's linear equations [Hås01]. This improved the bound for Max-4-Cut to about 0.987, but for Max-3-Cut their gadgets give only $55/57 \approx 0.965$, short of $16/17$. Table 1 collects these bounds.

1.1 Our results

Our first result improves the $16/17$ bound for Max-3-Cut.

Theorem 1.1. *For every $\varepsilon > 0$ it is NP-hard, given a weighted graph G, to distinguish between the case that some 3-colouring cuts at least a $(29/30 - \varepsilon)$ fraction of the total edge weight and the case that every 3-colouring cuts at most a $(9/10 + \varepsilon)$ fraction. In particular,* Max-3-Cut *is NP-hard to approximate within $27/29 + \varepsilon \approx 0.931$.*

The same approach, with a few extra vertices added to the gadgets, gives improved bounds for every k.

Theorem 1.2. *For every $k \geq 3$ and $\varepsilon > 0$,* Max-k-Cut *is NP-hard to approximate within $1 - 2/\Gamma_{k-3} + \varepsilon$, where $\Gamma_L = 29 + 18L + 3\binom{L}{2}$. In particular,* Max-4-Cut *is NP-hard to approximate within $45/47 + \varepsilon \approx 0.957$, and* Max-5-Cut *within $33/34 + \varepsilon \approx 0.971$.*

For $k = 3$, Theorem 1.2 is Theorem 1.1. Unlike the result of [AOTW14], our reductions do not have perfect completeness: on 3-colourable graphs, $16/17$ remains the best known bound. For k-colourable graphs with $k \geq 4$, however, we observe that the bounds previously stated in the literature can also be improved, by applying a known reduction more carefully.

Theorem 1.3. *For every $\varepsilon > 0$ it is NP-hard to distinguish 4-colourable graphs from graphs in which every 4-colouring cuts at most an $82/85 + \varepsilon \approx 0.965$ fraction of the edge weight, and 5-colourable graphs from graphs in which every 5-colouring cuts at most a $119/122 + \varepsilon \approx 0.975$ fraction. Similar improvements hold for every k from 4 to 20 (Section 6).*

Table 1 and Fig. 1 compare our bounds with the previous ones. Together, Theorems 1.2 and 1.3 improve the best known NP-hardness of Max-k-Cut for every k from 3 to 20; for larger k the bound of [AOTW14] remains the best. All three theorems have short proofs by hand.

	on k-colourable graphs		on general graphs	
	previous [AOTW14]	Theorem 1.3	previous	Theorems 1.1 and 1.2
$k=3$	$16/17 \approx 0.941$	—	16/17 [AOTW14]	$27/29 \approx 0.931$
$k=4$	$85/86 \approx 0.988$	$82/85 \approx 0.965$	0.987 [NRT25]	$45/47 \approx 0.957$
$k=5$	$120/121 \approx 0.992$	$119/122 \approx 0.975$	120/121 [AOTW14]	$33/34 \approx 0.971$
$k=6$	$101/102 \approx 0.990$	$53/54 \approx 0.981$	101/102 [AOTW14]	$45/46 \approx 0.978$
$k=7$	$136/137 \approx 0.993$	$202/205 \approx 0.985$	136/137 [AOTW14]	$117/119 \approx 0.983$

Table 1: NP-hardness of approximation for Max-k-Cut; smaller factors are stronger. The previous values for $k \geq 4$ are instances of the general formula $1 - \frac{1}{17(k+c_k)+c_k}$, $c_k = k \bmod 3$, of [AOTW14, Thm. 1.2], except the value 0.987 of [NRT25].

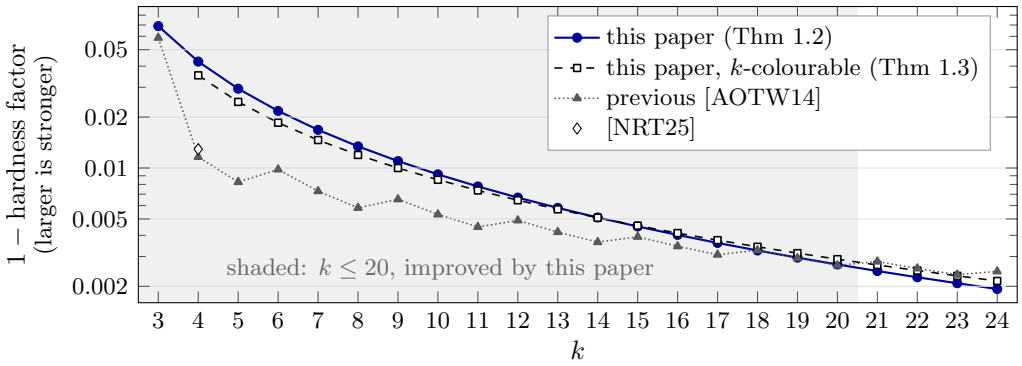


Figure 1: NP-hardness of approximation for Max-k-Cut, $3 \leq k \leq 24$, plotted as $1 -$ (hardness factor) on a logarithmic scale, so higher points are stronger bounds. The plotted values are $2/\Gamma_{k-3}$ for Theorem 1.2 (general graphs), $3/(51+34L+3\binom{L}{2})$ with $L = k-3$ for Theorem 1.3 (k-colourable graphs, $k \geq 4$), and $1/(17(k+c_k)+c_k)$ with $c_k = k \bmod 3$ for the previous bound of [AOTW14] (1/17 for $k = 3$); the diamond is the bound 0.987 of [NRT25] for Max-4-Cut. Theorem 1.2 is the strongest of the three for $k \leq 14$, Theorem 1.3 for $15 \leq k \leq 20$, and [AOTW14] for $k \geq 21$; in the shaded range $k \leq 20$ this paper improves the best previous bound.

1.2 Overview of the techniques

Gadget reductions and the colour-reference problem. Håstad [Hås01] proved that for systems of equations $x_1+x_2+x_3 = t$ over $\mathbb{Z}_3$ it is NP-hard to distinguish almost satisfiable systems from systems in which every assignment satisfies at most a $1/3+\varepsilon$ fraction of the equations. The classical way to turn this into hardness for a graph problem, introduced by Trevisan, Sorkin, Sudan and Williamson [TSSW00] for $\mathbb{Z}_2$ and used for $\mathbb{Z}_3$ by [NRT25], replaces each equation by a small graph, a *gadget* (Fig. 2), whose best cut is larger when the equation is satisfied. For $t \neq 0$, however, the predicate "$x_1 + x_2 + x_3 = t$" is not invariant under renaming the colours, so its gadget must refer to a fixed set of *global* vertices that play the role of the colours $0, 1, 2$. In a cheating colouring all global vertices may receive the same colour, and guarding against this makes the gadgets for $t \neq 0$ large and inefficient; in the analysis of [NRT25] they are the bottleneck that keeps the bound above 16/17.

The negation trick. Our main observation is that the negation $-X$ of an assignment X is again an assignment, and that $-X$ satisfies an equation with right-hand side t exactly when X satisfies the same equation with right-hand side $-t$. If every assignment satisfies at most a $1/3+\varepsilon$ fraction of

the equations, then applying this to both X and $-X$ shows that, for a random equation, the sum s of its variables under X satisfies $s = t$ or $s = -t$ with probability at most about $2/3$. It is therefore enough for the gadgets of the equations with $t = \pm 1$ to detect whether $s \neq 0$, which simply means that exactly two of the three variables are equal (Fig. 3). This predicate is invariant under all permutations of the colours, needs no global vertices, and is detected by a triangle together with one extra vertex (the gadget N of Fig. 4(b)). Combined with the gadget of [NRT25] for $t = 0$, this proves Theorem 1.1 in a few lines (Section 3).

A soundness criterion. For $k \geq 4$ the satisfying colourings use only three of the k colours, and global vertices become necessary again. We prove a general criterion for gadget reductions from Håstad's equations (Theorem 4.1): under a mild symmetry condition, the gadgets give NP-hardness of approximation within $\rho + \delta$ for every $\delta > 0$, where

$$\rho \;=\; \frac{\max_{x,y} \sum_t V_t(x,y)}{\sum_t C_t},$$

$V_t(x, y)$ is the value of the gadget for right-hand side t when its three variables receive the colours x and the global vertices the colours y, and C_t is its least value on satisfying colourings. In words, a cheating colouring can do no better than to use a single colour pattern x on all equations, and each pattern is charged with the sum of the values of all three gadgets. The proof decodes a colouring of the graph into many assignments of Håstad's instance and uses a short Fourier-analytic argument; it relies on the fact that Håstad's equations have one variable of one kind and two of another. We also show that no analysis that uses Håstad's theorem only as a black box can do better (Proposition 4.3). The criterion contains the negation trick and, for $k = 3$, the analysis of [NRT25] as special cases. It also turns gadget design into a single linear program over the three gadgets. Solving this program for $k = 4, 5$ and simplifying the solutions by hand led to the gadgets behind Theorem 1.2: the gadgets of Theorem 1.1 together with $k - 3$ global vertices of the extra colours (Section 5).

Universal vertices. Theorem 1.3 uses a reduction of [KKLP97, GS13] that adds vertices joined to the whole graph and forces them to take the extra colours. The previously stated constants apply it on top of a general bound for K-colourable graphs (K divisible by 3) which, for $K = 3$, is three times weaker than the $16/17$ bound it is derived from; applying it directly to the $16/17$ bound gives Theorem 1.3 (Section 6). Placing the extra vertices inside the gadgets, as in Theorem 1.2, is more economical than adding them to the final graph.

A barrier. Finally, we show that $27/29$ is the best factor that the criterion can give for Max-3-Cut, for gadgets of any size (Theorem 7.1): there is a single distribution of colour patterns on which every gadget, for every right-hand side, still achieves at least a $27/29$ fraction of its completeness value on average (Fig. 5). The proof uses LP duality, in the spirit of the optimality proofs of [TSSW00, HHM$^+$17], with explicit rational certificates that we verify by computer in exact arithmetic; this is the only computer-assisted result of the paper. To go beyond $27/29$, one has to change the source problem (for instance, to other predicates covered by Chan's theorem [Cha16]), the form of the reduction, or the analysis.

How these results were obtained. This research was carried out with extensive use of an AI system, which designed and ran the computer searches, found the arguments and drafted the paper under the direction of the author. The methodology, including how every claim was checked, is described in Appendix B and in the AI methodology statement at the end of the paper.

Organisation. Section 2 fixes notation and states Håstad's theorem in the form we use. Section 3 proves Theorem 1.1, Section 4 the soundness criterion, and Section 5 Theorem 1.2. Section 6 treats universal vertices and Theorem 1.3, Section 7 the barrier, and Section 8 open problems. Appendix A collects further remarks on the soundness criterion, and Appendix B describes the computations.

2 Preliminaries

Cuts. We write $[k] = \{0, 1, \dots, k-1\}$ and identify $[3]$ with $\mathbb{Z}_3$. A (weighted) graph $G = (V, E, w)$ has non-negative edge weights; for a colouring $\chi\colon V \to [k]$ let $\mathrm{cut}_\chi(G) = \sum_{\{u,v\}\in E} w_{uv}\,[\chi(u) \neq \chi(v)]$ and $w(G) = \sum_{\{u,v\}\in E} w_{uv}$. We say that Max-$k$-Cut is (c, s)-hard if it is NP-hard to distinguish $\max_\chi \mathrm{cut}_\chi(G) \geq c\,w(G)$ from $\max_\chi \mathrm{cut}_\chi(G) \leq s\,w(G)$; this implies NP-hardness of approximation within s/c. Weights may be removed at an arbitrarily small loss by the method of [CST01]; see [GS13, Remark 2.6] for Max-k-Cut. The transformation replaces vertices by independent sets and edges by bipartite graphs, so it also preserves k-colourability; hence all results hold for unweighted simple graphs.

Håstad's equations. A *role-separated* instance Φ of $3\textsc{Lin}(\mathbb{Z}_3)$ consists of disjoint sets U and W of variables and a collection of equations e of the form $u_e + w_e + w'_e = t_e$ over $\mathbb{Z}_3$, with $u_e \in U$, distinct $w_e, w'_e \in W$, $t_e \in \mathbb{Z}_3$ (the *type* of e), and weights $\lambda_e \geq 0$ summing to 1. For an assignment $\sigma\colon U \cup W \to \mathbb{Z}_3$, $\mathrm{val}_\Phi(\sigma)$ is the total weight of the equations it satisfies, and $\mathrm{val}(\Phi) = \max_\sigma \mathrm{val}_\Phi(\sigma)$. Φ is *balanced* if each type has total weight $1/3$.

Theorem 2.1 (Håstad [Hås01])**.** *For every $\varepsilon > 0$ it is NP-hard, given a balanced role-separated instance Φ of $3\textsc{Lin}(\mathbb{Z}_3)$, to distinguish $\mathrm{val}(\Phi) \geq 1 - \varepsilon$ from $\mathrm{val}(\Phi) \leq 1/3 + \varepsilon$.*

Comments on the statement. This is Håstad's theorem for the group $\mathbb{Z}_3$ [Hås01, Thm. 5.9]; the gap version follows from Lemmas 5.10 and 5.11 there. In his verifier, each test reads one entry of a folded long-code table indexed by a set U of variables and two entries of a table indexed by a set W of clauses, and accepts iff the three values sum to a constant; folding only changes the right-hand side. Giving the two provers separate tables makes the instance role-separated; the soundness analysis [Hås01, Lemma 5.11] applies to arbitrary tables, i.e. to arbitrary assignments, and constructs the strategies of the two provers from the two kinds of tables separately. Tests whose two W-entries fall into the same folded class have negligible total weight (it tends to 0 with the parameter of the verifier); we discard them, which changes completeness and soundness by a negligible amount.

To balance the types, replace every $u \in U$ by three copies $u^{(0)}, u^{(1)}, u^{(2)}$ and each equation e by the three equations $u_e^{(c)} + w_e + w'_e = t_e + c$ ($c \in \mathbb{Z}_3$), each of weight $\lambda_e/3$. The new instance Φ' is role-separated (with $U \times \mathbb{Z}_3$ and W), and every type has weight exactly $1/3$. An assignment σ of Φ yields $\sigma(u^{(c)}) = \sigma(u) + c$, which satisfies the same weight. Conversely, for any assignment τ of Φ',

$$\mathrm{val}_{\Phi'}(\tau) = \tfrac{1}{3}\textstyle\sum_{c\in\mathbb{Z}_3} \mathrm{val}_\Phi(\sigma_c), \qquad \sigma_c(u) = \tau(u^{(c)}) - c, \quad \sigma_c(w) = \tau(w),$$

and each σ_c is an assignment of Φ (the values on W are shared), so $\mathrm{val}(\Phi') \leq \mathrm{val}(\Phi)$. □

Gadgets. Fix $k \geq 3$. A *gadget* I is a weighted graph on three *primary* vertices p_1, p_2, p_3, k *global* vertices $g_0, \dots, g_{k-1}$ and some *auxiliary* vertices. For $x \in [k]^3$ and $y \in [k]^k$ let

$$V_I(x, y) = \max\{\mathrm{cut}_\chi(I) : \chi(p_j) = x_j,\ \chi(g_i) = y_i\},$$

the maximum being over the colours of the auxiliary vertices. Let $y^\circ = (0, 1, \ldots, k-1)$ be the canonical colouring of the globals. The value of a colouring does not change when the k colours are permuted, so we may normalise the colours of the globals: let $\mathcal{Y}_k \subseteq [k]^k$ contain one representative of each orbit under permutations of the colours (we use restricted growth strings: the first entry is 0, and each entry exceeds the maximum of the previous ones by at most 1). Since $V_I(\pi x, \pi y) = V_I(x, y)$ for every permutation π of $[k]$, maxima of V_I, or of sums of several V_I, jointly over $x \in [k]^3$ and $y \in \mathcal{Y}_k$ equal the maxima over $x \in [k]^3$ and all $y \in [k]^k$; the normalisation is only a computational saving. A *family* for $3\textsc{Lin}(\mathbb{Z}_3)$ is a triple (I_0, I_1, I_2) of gadgets, I_t to be used for equations of type t. We write $V_t = V_{I_t}$ and $S_y(x) = \sum_t V_t(x, y)$. Let $\mathrm{sat}_t = \{x \in \mathbb{Z}_3^3 : x_1 + x_2 + x_3 = t\}$ be the set of *satisfying* tuples of type t; they use only the colours $0, 1, 2$. The *completeness* of I_t is

$$C_t = \min\{V_t(x, y^\circ) : x \in \mathrm{sat}_t\}.$$

The reduction. Given a role-separated Φ and a family, the graph $J = J(\Phi)$ (Fig. 2) has a vertex for every variable of Φ, k global vertices, and for every equation e a copy of I_{t_e} scaled by λ_e, whose primaries p_1, p_2, p_3 are identified with u_e, w_e, w'_e, whose global vertices are identified with the global vertices of J, and whose auxiliary vertices are new. For balanced Φ, $w(J) = \frac{1}{3}\sum_t w(I_t)$. If $\mathrm{val}(\Phi) \geq 1 - \varepsilon$ and Φ is balanced, colouring the variables by an assignment that satisfies weight at least $1-\varepsilon$, the globals by y°, and the auxiliary vertices of each satisfied equation optimally gives

$$\max_\chi \mathrm{cut}_\chi(J) \;\geq\; \tfrac{1}{3}\textstyle\sum_t C_t \;-\; \varepsilon \max_t C_t. \tag{1}$$

For every colouring χ of J, after normalising $y = \chi(g_0, \ldots, g_{k-1}) \in \mathcal{Y}_k$,

$$\mathrm{cut}_\chi(J) \;\leq\; \sum_e \lambda_e \, V_{t_e}\big(\chi(u_e), \chi(w_e), \chi(w'_e),\, y\big), \tag{2}$$

since the auxiliary vertices of different copies are disjoint.

3 Max-3-Cut: the negation trick

Here $k = 3$, and the gadgets do not use their global vertices at all. Write $s(x) = x_1 + x_2 + x_3 \in \mathbb{Z}_3$ and observe that $s(x) = 0$ iff the three colours are all equal or all distinct, and $s(x) \neq 0$ iff exactly two of them are equal (Fig. 3).

- The *E-gadget* E (Fig. 4(a)) is the type-0 gadget of [NRT25, Lemma C.1]: auxiliary vertices $a_1, \ldots, a_6$ and the 18 unit-weight edges

  $$p_1a_1,\ p_1a_2,\ p_1a_3,\ p_1a_4,\quad p_2a_3,\ p_2a_4,\ p_2a_5,\ p_2a_6,\quad p_3a_1,\ p_3a_2,\ p_3a_5,\ p_3a_6,$$
  $$a_1a_6,\ a_1a_4,\ a_2a_3,\ a_2a_5,\ a_3a_6,\ a_4a_5.$$

  The auxiliary vertices form the 6-cycle $a_1a_4a_5a_2a_3a_6$, each is adjacent to two primaries, and opposite vertices of the cycle are adjacent to the same pair of primaries; hence E is invariant under all permutations of the primaries.

- The *N-gadget* N (Fig. 4) consists of a unit-weight triangle on p_1, p_2, p_3 and one auxiliary vertex q joined to p_1, p_2, p_3 by edges of weight 3 (total weight 12).

Lemma 3.1. *For all $x \in \mathbb{Z}_3^3$: $V_E(x) = 18$ if $s(x) = 0$ and $V_E(x) = 16$ otherwise; $V_N(x) = 11$ if $s(x) \neq 0$ and $V_N(x) = 9$ otherwise.*

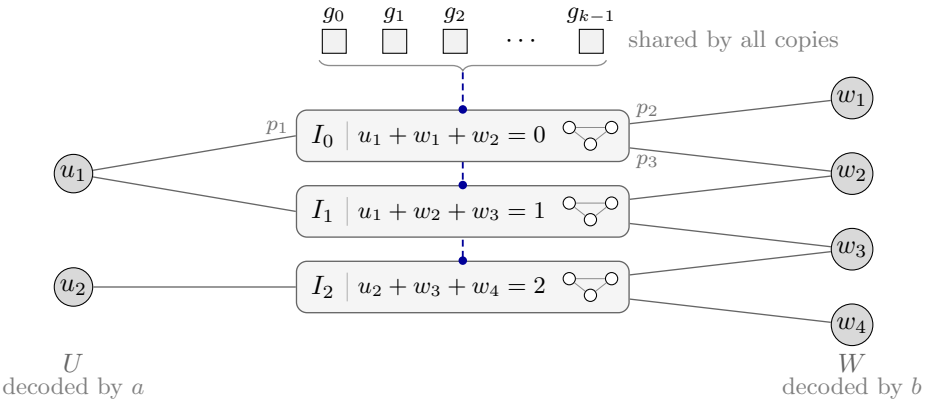


Figure 2: The graph $J(\Phi)$ for a balanced role-separated instance Φ with three equations of weight $\frac{1}{3}$, one of each type. Each variable is a single vertex, shared by the equations that contain it (u_1, w_2 and w_3 occur twice). Each equation e contributes a copy of I_{t_e} scaled by λ_e (box) with new auxiliary vertices (small circles, schematic); its primaries p_1, p_2, p_3 (labelled on the top copy) are identified with u_e, w_e, w'_e (grey lines, which mark identifications and are not edges of J) and its globals with the shared vertices $g_0, \dots, g_{k-1}$ (dashed line, drawn behind the boxes), so different copies meet only in variable vertices and globals. Section 4 decodes a colouring χ into the assignment $u \mapsto a(\chi(u))$ on U, $w \mapsto b(\chi(w))$ on W, for arbitrary maps $a, b\colon [k] \to \mathbb{Z}_3$.

Proof. By symmetry under permutations of the primaries and of the colours, it suffices to consider $x = (0,0,0)$, $x = (0,1,2)$ and $x = (0,0,1)$. For $(0,0,0)$, colour the cycle alternately with 1 and 2; for $(0,1,2)$, give each a_j the colour of the primary it is not adjacent to (consecutive vertices of the cycle then differ). In both cases all 18 edges are cut. For $x = (0,0,1)$, each of a_1, a_2, a_5, a_6 is adjacent to a primary of colour 0 and one of colour 1, so it has an uncut edge unless it has colour 2; since a_1a_6 and a_2a_5 are disjoint edges, at least two edges are uncut. Colouring the cycle $a_1a_4a_5a_2a_3a_6$ by $2,1,2,1,2,1$ leaves exactly p_3a_2 and p_3a_6 uncut, so $V_E(x) = 16$. (This agrees with [NRT25, Lemma C.1], which states completeness 18 and value at most 16 on unsatisfying tuples.)

For N: if exactly two of the x_j are equal, the triangle contributes 2 and q can take the third colour, contributing 9. If all are equal, the triangle contributes 0 and q contributes 9. If all are distinct, the triangle contributes 3 and q agrees with one primary, contributing 6. □

Proof of Theorem 1.1. Use the family $(2E, N, N)$, i.e. E with all weights doubled for type 0 and N for types 1, 2. By Lemma 3.1, for every x and every type t,

$$V_t(x) \;=\; \beta_t + 2\big([s(x) = t] + [s(x) = -t]\big), \qquad \beta_0 = 32,\ \beta_1 = \beta_2 = 9. \tag{3}$$

Indeed, for $t = 0$ the bracket is $2[s = 0]$, and for $t \neq 0$ it is $[s \neq 0]$. Let Φ be an instance from Theorem 2.1 and $J = J(\Phi)$, so $w(J) = \frac{1}{3}(36 + 12 + 12) = 20$.

Completeness: by (1), $\max_\chi \mathrm{cut}_\chi(J) \geq \frac{1}{3}(36 + 11 + 11) - 36\varepsilon = \frac{58}{3} - 36\varepsilon = (\frac{29}{30} - \frac{9}{5}\varepsilon)\, w(J)$.

Soundness: let χ be any colouring and σ the assignment it induces on the variables of Φ. By (2) and (3),

$$\mathrm{cut}_\chi(J) \leq \sum_e \lambda_e \beta_{t_e} + 2\sum_e \lambda_e\big([s_e = t_e] + [s_e = -t_e]\big) = \tfrac{50}{3} + 2\big(\mathrm{val}_\Phi(\sigma) + \mathrm{val}_\Phi(-\sigma)\big),$$

where $s_e = \sigma(u_e) + \sigma(w_e) + \sigma(w'_e)$, because $-\sigma$ satisfies e iff $-s_e = t_e$. If $\mathrm{val}(\Phi) \leq \frac{1}{3} + \varepsilon$, both values are at most $\frac{1}{3} + \varepsilon$, so $\mathrm{cut}_\chi(J) \leq \frac{50}{3} + \frac{4}{3} + 4\varepsilon = 18 + 4\varepsilon = (\frac{9}{10} + \frac{\varepsilon}{5})\, w(J)$. □

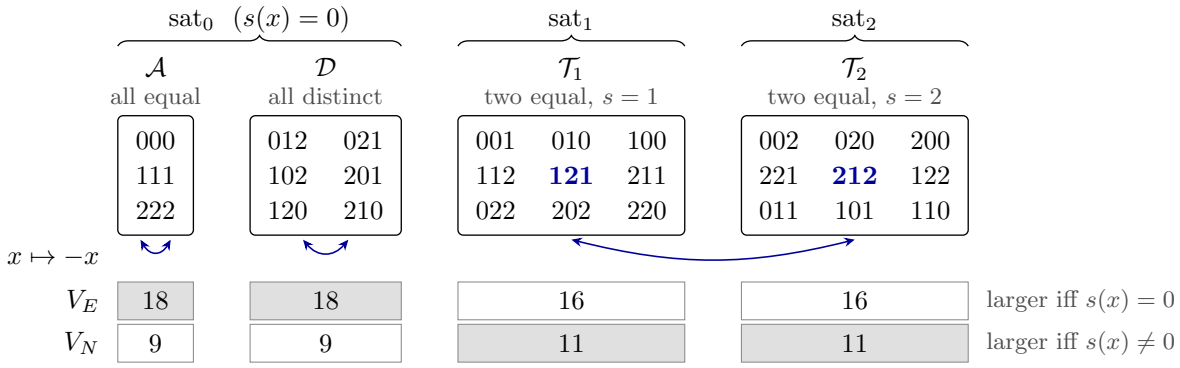


Figure 3: The 27 tuples $x = x_1x_2x_3 \in \mathbb{Z}_3^3$ in the classes $\mathcal{A}$ (all equal), $\mathcal{D}$ (all distinct), $\mathcal{T}_1$, $\mathcal{T}_2$ (two equal, $s(x) = x_1 + x_2 + x_3 = 1$ resp. 2); the tuples with $s(x) = t$ form sat$_t$, so sat$_0 = \mathcal{A} \cup \mathcal{D}$, sat$_1 = \mathcal{T}_1$ and sat$_2 = \mathcal{T}_2$. Negation $x \mapsto -x$ maps $\mathcal{A}$ and $\mathcal{D}$ to themselves (on $\mathcal{D}$ it exchanges the two columns) and maps each tuple of $\mathcal{T}_1$ to the tuple of $\mathcal{T}_2$ at the same position (e.g. $121 \mapsto 212$). The bottom rows give the gadget values of Lemma 3.1, with the larger value of each gadget shaded: V_E is larger exactly on sat$_0$, and V_N exactly on sat$_1 \cup$ sat$_2$, so N cannot tell $\mathcal{T}_1$ from $\mathcal{T}_2$.

Remark 3.2. The proof of Theorem 1.1 uses neither role separation nor, essentially, balance: if the type-0 equations have total weight λ_0 (and types 1, 2 together $1 - \lambda_0$), the same computation gives the ratio $(23\lambda_0 + \frac{31}{3})/(25\lambda_0 + 11) \leq 31/33 < 16/17$. (For role-separated instances the soundness hypothesis even forces balance in the NO case, see Remark A.2.)

Remark 3.3. In the per-type framework of [NRT25, TSSW00], N is useless as a type-1 gadget: it does not distinguish $s = 1$ from $s = 2$, so its value on unsatisfying inputs can equal its completeness. The gain comes entirely from the analysis, which pairs σ with $-\sigma$.

4 A soundness criterion for gadgets from Håstad's equations

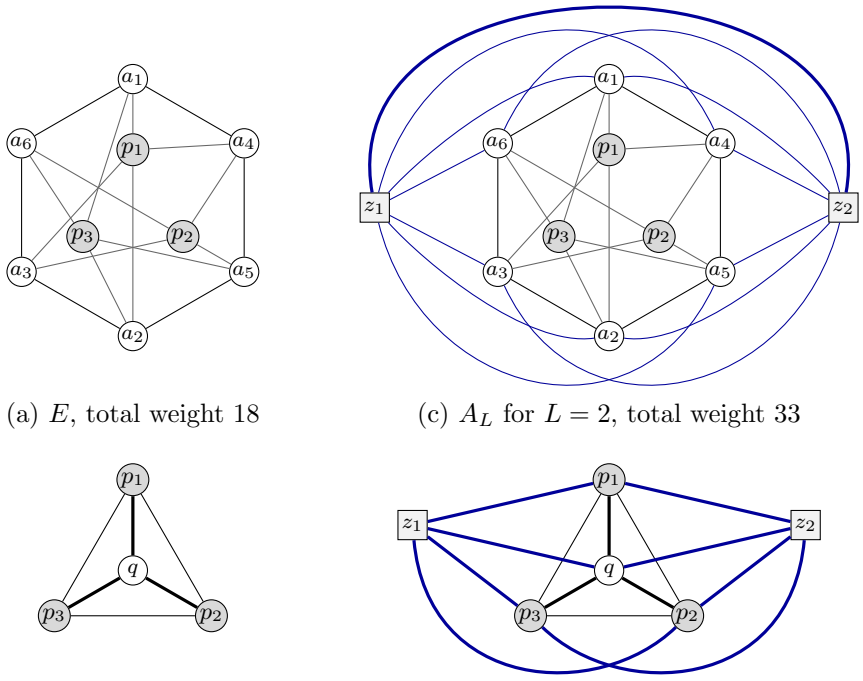
This section proves the criterion behind Theorems 1.2 and 7.1. Readers interested only in Max-3-Cut may skip it: Theorem 1.1 was proved directly in Section 3, and Sections 5 and 7 use only the statement of Theorem 4.1. Throughout, $k \geq 3$.

Theorem 4.1 (Soundness criterion)**.** *Let (I_0, I_1, I_2) be a family of gadgets for* Max-k-Cut*, $k \geq 3$, such that each $V_t(x, y)$ is symmetric under exchanging x_2 and x_3. Let*

$$\rho = \frac{\max_{y \in \mathcal{Y}_k} \max_{x \in [k]^3} S_y(x)}{\sum_{t \in \mathbb{Z}_3} C_t}.$$

Then for every $\delta > 0$, Max-k-Cut *is NP-hard to approximate within $\rho + \delta$. More precisely, with $\bar{w} = \sum_t w(I_t)$,* Max-k-Cut *is $\big(\frac{\sum_t C_t}{\bar{w}} - \delta,\ \frac{\max_{x,y} S_y(x)}{\bar{w}} + \delta\big)$-hard.*

Proof idea. Fix a colouring χ of $J(\Phi)$, and let $P(t, x)$ be the weight of the type-t equations whose three variables receive the colours $x \in [k]^3$. By (2), the cut is at most $\sum_{t,x} P(t, x)V_t(x, y)$. If, under P, the type t were uniform and independent of the colour pattern x, this would be an average of $\frac{1}{3}S_y(x)$ over patterns, hence at most $\frac{1}{3}\max_x S_y(x)$. To compare P with such a distribution, we decode χ into assignments of Φ: for any two maps $a, b\colon [k] \to \mathbb{Z}_3$, a U-variable of colour c gets the value $a(c)$ and a W-variable of colour c gets the value $b(c)$ (Fig. 2). In the NO case val$(\Phi) \leq \frac{1}{3} + \varepsilon$



(a) E, total weight 18

(c) A_L for $L = 2$, total weight 33

(b) N, total weight 12

(d) N_L for $L = 2$, total weight 36

Figure 4: Left: the gadgets of Theorem 1.1; right: their extensions for Max-k-Cut with $k = 3 + L$ (Section 5), drawn for $L = 2$. Thin edges have weight 1 and very thick edges weight 3; primaries are grey, the universal vertices $z_1, \dots, z_L$ (the globals $g_3, \dots, g_{k-1}$) are squares, and edges at universal vertices are blue. (a) The type-0 gadget E of [NRT25], whose auxiliary vertices form the 6-cycle $a_1a_4a_5a_2a_3a_6$; (b) the gadget N for types 1 and 2; (c) A_L adds an edge of weight 1 from every a_j to every z_i (none from the primaries) and one of weight 3 between any two z_i; (d) N_L adds edges of weight 3 from every z_i to q and to every primary, and none between the z_i.

of Theorem 2.1, each decoded assignment satisfies at most $\frac{1}{3} + \varepsilon$, and hence (shifting a) at least $\frac{1}{3} - 2\varepsilon$, of the equations, so every test $g_{a,b}$ defined below sees t as nearly uniform and independent of x. The spanning lemma (Lemma 4.2) shows that these tests detect every dependence between t and x that can affect the cut.

Two features of the decoding explain the hypotheses. The two W-variables of an equation are decoded by the same map b, so the tests cannot tell x_2 from x_3; this is why V_t must be symmetric in them (condition (ii) of Lemma 4.2). The U- and W-variables must be decoded by independent maps: the proof of Lemma 4.2 shifts a alone by a constant c, which changes the decoded sum by c. If U and W could overlap, only maps with $a = b$ would be available, a common shift would change the sum by $3c = 0$, and the argument would break down. This is why we need role-separated instances.

The tests. Let $\omega = e^{2\pi i/3}$. For maps $a, b\colon [k] \to \mathbb{Z}_3$ define $g_{a,b}\colon \mathbb{Z}_3 \times [k]^3 \to \{0, 1\}$ by

$$g_{a,b}(t, x) = [\, a(x_1) + b(x_2) + b(x_3) = t \,].$$

Thus $g_{a,b}(t, x) = 1$ exactly when an equation of type t whose variables u_e, w_e, w'_e have the colours x_1, x_2, x_3 is satisfied by the decoded assignment $u \mapsto a(\chi(u))$, $w \mapsto b(\chi(w))$.

Lemma 4.2 (Spanning lemma)**.** *Let $f\colon \mathbb{Z}_3\times[k]^3\to\mathbb{R}$ satisfy (i) $\sum_{t\in\mathbb{Z}_3} f(t,x)=0$ for every x, and (ii) $f(t,x_1,x_2,x_3)=f(t,x_1,x_3,x_2)$. Then f is a real linear combination of the functions $g_{a,b}-\frac13$.*

Proof. We expand f in the characters $\omega^{\ell t}$ of the variable t. By (i) the coefficient of ω^0 vanishes, and by (ii) the coefficients of ω^t and ω^{2t} are functions of x in $\mathbb{C}^k\otimes\mathrm{Sym}^2(\mathbb{C}^k)$. We show that the products $A(x_1)B(x_2)B(x_3)$ that arise from the $g_{a,b}$ span this space.

We work over $\mathbb{C}$. For $c\in\mathbb{Z}_3$ let $a{+}c$ denote the map $v\mapsto a(v)+c$. For fixed (t,x), $g_{a+c,b}(t,x)=1$ for exactly one c, namely $c=d:=t-a(x_1)-b(x_2)-b(x_3)$. Hence for $\ell\in\{1,2\}$,

$$h_{\ell,a,b}:=\sum_{c\in\mathbb{Z}_3}\omega^{\ell c}\big(g_{a+c,b}-\tfrac13\big)=\omega^{\ell d}=\omega^{\ell t}\,\omega^{-\ell a(x_1)}\,\omega^{-\ell b(x_2)}\,\omega^{-\ell b(x_3)}.$$

Let $\mathcal{R}\subseteq\mathbb{C}^k$ be the set of vectors whose entries are cube roots of unity. As a and b range over all maps, $\omega^{-\ell a(\cdot)}$ and $\omega^{-\ell b(\cdot)}$ range over all of $\mathcal{R}$ (for both $\ell=1,2$). Thus the span contains $\omega^{\ell t}A(x_1)B(x_2)B(x_3)$ for all $A,B\in\mathcal{R}$.

$\mathcal{R}$ spans $\mathbb{C}^k$: it contains $\mathbf{1}$ and $\mathbf{1}+(\omega-1)e_i$. The tensors $B\otimes B$, $B\in\mathcal{R}$, span the symmetric square $\mathrm{Sym}^2(\mathbb{C}^k)$: if a symmetric matrix Q satisfies $\sum_{i,j}Q_{ij}B_iB_j=0$ for all $B\in\mathcal{R}$, then writing $B_i=\omega^{\theta_i}$ this says $\sum_{i\le j}Q'_{ij}\,\omega^{\theta_i+\theta_j}=0$ for all $\theta\in\mathbb{Z}_3^k$, with $Q'_{ii}=Q_{ii}$ and $Q'_{ij}=2Q_{ij}$ for $i<j$. The functions $\theta\mapsto\omega^{\theta_i+\theta_j}$ $(i\le j)$ are distinct characters of $\mathbb{Z}_3^k$ (their exponent vectors e_i+e_j are distinct since $2\not\equiv 0,1 \pmod 3$), hence linearly independent, so $Q=0$. Consequently the span contains $\omega^{\ell t}\varphi(x_1)\psi(x_2,x_3)$ for every $\varphi\in\mathbb{C}^k$, every symmetric $\psi\in\mathrm{Sym}^2(\mathbb{C}^k)$ and $\ell=1,2$.

Now expand $f(t,x)=\sum_{\ell\in\mathbb{Z}_3}\omega^{\ell t}\hat f_\ell(x)$ with $\hat f_\ell(x)=\frac13\sum_t\omega^{-\ell t}f(t,x)$. Condition (i) gives $\hat f_0=0$, and (ii) makes $\hat f_1,\hat f_2$ symmetric in (x_2,x_3), i.e. elements of $\mathbb{C}^k\otimes\mathrm{Sym}^2(\mathbb{C}^k)$. So f lies in the complex span of the $g_{a,b}-\frac13$; since f and the $g_{a,b}$ are real, taking real parts of the coefficients gives a real combination. □

Proof of Theorem 4.1. Reduce from Theorem 2.1 as in Section 2; then $w(J)=\bar w/3$. Completeness is (1). For soundness, fix $y\in\mathcal{Y}_k$. The function $f_y(t,x)=V_t(x,y)-\frac13 S_y(x)$ satisfies (i) and (ii) of Lemma 4.2, so $f_y=\sum_{a,b}\gamma^y_{a,b}(g_{a,b}-\frac13)$ for some reals $\gamma^y_{a,b}$. Let $K_\gamma=\max_y\sum_{a,b}|\gamma^y_{a,b}|$; it depends only on the family.

Let Φ satisfy $\mathrm{val}(\Phi)\le\frac13+\varepsilon$, let χ be a colouring of $J(\Phi)$, and normalise so that the globals have colours $y\in\mathcal{Y}_k$. Let $P(t,x)$ be the total weight of the equations e with $t_e=t$ and $(\chi(u_e),\chi(w_e),\chi(w'_e))=x$, and let $\bar P(x)=\sum_t P(t,x)$; both are probability distributions, since the weights λ_e sum to 1. For any maps $a,b\colon[k]\to\mathbb{Z}_3$, the decoded assignment $\sigma_{a,b}(u)=a(\chi(u))$ for $u\in U$ and $\sigma_{a,b}(w)=b(\chi(w))$ for $w\in W$ is well defined because $U\cap W=\emptyset$, and it satisfies exactly the weight $\langle P,g_{a,b}\rangle$. Hence $\langle P,g_{a,b}\rangle\le\frac13+\varepsilon$. Since $\sum_c g_{a+c,b}\equiv 1$, also $\langle P,g_{a,b}\rangle=1-\sum_{c\neq0}\langle P,g_{a+c,b}\rangle\ge\frac13-2\varepsilon$. Therefore

$$\begin{aligned}
\mathrm{cut}_\chi(J)&\le\sum_{t,x}P(t,x)\,V_t(x,y) && \text{by (2)},\\
&=\sum_x\bar P(x)\,\frac{S_y(x)}{3}+\sum_{a,b}\gamma^y_{a,b}\Big\langle P,\,g_{a,b}-\tfrac13\Big\rangle && \text{as } V_t=\tfrac13 S_y+f_y, \text{ expanding } f_y,\\
&=\sum_x\bar P(x)\,\frac{S_y(x)}{3}+\sum_{a,b}\gamma^y_{a,b}\Big(\langle P,g_{a,b}\rangle-\tfrac13\Big) && \text{as } \textstyle\sum_{t,x}P(t,x)=1,\\
&\le\frac13\max_x S_y(x)+2K_\gamma\varepsilon && \text{as } \langle P,g_{a,b}\rangle\in[\tfrac13-2\varepsilon,\tfrac13+\varepsilon].
\end{aligned}$$

Choosing ε small enough compared to δ finishes the proof. □

Tightness. The numerator of ρ cannot be improved by any analysis that uses Håstad's theorem as a black box.

Proposition 4.3. *For every family, every $x^* \in [k]^3$ and every $y^* \in [k]^k$, there is a balanced role-separated instance Φ_0 with $\mathrm{val}_{\Phi_0}(\sigma) = \frac{1}{3}$ for every assignment σ, and a colouring of $J(\Phi_0)$ of value $\frac{1}{3}S_{y^*}(x^*)$. Hence no soundness analysis that uses only the NO conclusion of Theorem 2.1 can improve on the numerator of ρ.*

Proof. Let Φ_0 consist of one variable $u \in U$, two variables $w, w' \in W$ and the three equations $u + w + w' = t$, $t \in \mathbb{Z}_3$, of weight $\frac{1}{3}$ each (or a disjoint union of copies of it). Every assignment satisfies exactly one of the three equations. Colour u, w, w' by x^*_1, x^*_2, x^*_3, the globals by y^*, and the auxiliary vertices optimally. □

In the language of the proof of Theorem 4.1, Φ_0 realises the distribution $P(t, x) = \frac{1}{3}[x = x^*]$, which satisfies $\langle P, g_{a,b}\rangle = \frac{1}{3}$ for all a, b and so passes every test. Moreover, if the maximum in ρ is attained at some (x^*, y^*) with $x^*_2 = x^*_3$, then *every* $J(\Phi)$ with Φ balanced has a colouring of value $\frac{1}{3}\max_{x,y} S_y(x)$: colour all of U by x^*_1, all of W by x^*_2 and the globals by y^*. The analysis is then exact for every instance, not only for Φ_0. This is the case for all families in this paper, where the maximum is attained at a constant tuple.

Further properties. The criterion is consistent with the known algorithms: for every family, $\rho \geq \alpha_k$, where α_k is the approximation ratio of the Frieze–Jerrum rounding [FJ97, dKPW04] (Proposition A.4). Appendix A also shows that the symmetry hypothesis of Theorem 4.1 costs nothing (Remark A.1), that the theorem does not need balance in the NO case (Remark A.2), and that for $k = 3$ it is never worse than the per-type analysis of [NRT25] (Remark A.3).

5 A uniform family for Max-(3+L)-Cut

Let $L \geq 0$ and $k = 3 + L$. Call the global vertices $g_3, \ldots, g_{k-1}$ (colours $3, \ldots, k-1$ in y°) the *universal* vertices $z_1, \ldots, z_L$; the globals g_0, g_1, g_2 are not used (Fig. 4(c),(d)).

- A_L: the E-gadget of Section 3 (unit weights), plus an edge of weight 1 between every auxiliary vertex a_j and every universal vertex, plus an edge of weight 3 between every pair of universal vertices. Total weight $18 + 6L + 3\binom{L}{2}$.
- N_L: a unit-weight triangle on the primaries, one auxiliary vertex q joined with weight 3 to p_1, p_2, p_3 and to every universal vertex, and an edge of weight 3 between every primary and every universal vertex. Total weight $12 + 12L$.

The family is $(2A_L, N_L, N_L)$. For $L = 0$ it is the family of Section 3.

Lemma 5.1. *For every $L \geq 0$, the gadgets A_L and N_L are invariant under all permutations of the primaries, and the family $(2A_L, N_L, N_L)$ satisfies*

$$C_0 = w(2A_L), \qquad C_1 = C_2 = w(N_L) - 1, \qquad \max_{x,y} S_y(x) = \textstyle\sum_t C_t - 4.$$

Consequently $\sum_t C_t = 2\Gamma_L$ and $\sum_t w(I_t) = 2\Gamma_L + 2$, with $\Gamma_L = 29 + 18L + 3\binom{L}{2}$.

Proof. For a colouring of a gadget, write $\mathrm{col}(v)$ for the colour of a vertex v.

Completeness. Let $x \in \{0,1,2\}^3$ and let the globals have colours y°, so the universal vertices have distinct colours $3,\dots,k-1$, different from all colours of the primaries. For $t=0$, colour the a_j as in the proof of Lemma 3.1 (colours in $\{0,1,2\}$): all edges of E, all edges $a_j z_i$ and all edges $z_i z_{i'}$ are cut, so $C_0 = w(2A_L)$. For $t \neq 0$, exactly two primaries are equal; colour q by the third colour of $\{0,1,2\}$: every edge except one triangle edge is cut, so $C_1 = C_2 = w(N_L) - 1$.

Soundness. Fix $x \in [k]^3$ and y, and let $\ell_A = w(A_L) - V_{A_L}(x,y)$ and $\ell_N = w(N_L) - V_{N_L}(x,y)$ be the least uncut weights. Since $S_y(x) = 2V_{A_L} + 2V_{N_L} = \sum_t w(I_t) - 2(\ell_A + \ell_N) = \sum_t C_t + 2 - 2(\ell_A + \ell_N)$, it suffices to show $\ell_A + \ell_N \geq 3$.

In N_L the vertex q is optimally coloured with a colour that is least frequent among its neighbours, so ℓ_N equals the uncut weight of the triangle, plus 3 for every pair of a primary and a universal vertex of the same colour, plus $3 \min_c n_c$, where n_c is the number of neighbours of q (among $p_1, p_2, p_3, z_1, \dots, z_L$) of colour c. In A_L every pair $z_i z_{i'}$ of equal colour costs 3. Hence we may assume that no primary shares its colour with a universal vertex, that the z_i have distinct colours, and that some colour is missing among the $k = 3 + L$ neighbours of q; otherwise $\ell_A \geq 3$ or $\ell_N \geq 3$. If x is constant, the triangle gives $\ell_N = 3$. If x_1, x_2, x_3 are distinct, then the primaries and the z_i use all k colours, which we excluded. If exactly two coincide, then by symmetry (permuting the primaries and the colours) $x = (0,0,1)$; the triangle gives $\ell_N \geq 1$, and exactly one colour c^* lies outside $\{0, 1, \mathrm{col}(z_1), \dots, \mathrm{col}(z_L)\}$. Each of a_1, a_2, a_5, a_6 is adjacent to a primary of colour 0, to a primary of colour 1, and to all z_i, so it has an uncut edge unless it has colour c^*. Since $a_1 a_6$ and $a_2 a_5$ are disjoint edges of E, each of the pairs $\{a_1, a_6\}$ and $\{a_2, a_5\}$ contributes an uncut edge, so $\ell_A \geq 2$. In all cases $\ell_A + \ell_N \geq 3$.

Equality holds for constant x and distinct universal colours different from x_1: then $\ell_N = 3$, and the 6-cycle of E can be coloured with the two colours outside $\{x_1, \mathrm{col}(z_1), \dots, \mathrm{col}(z_L)\}$, so $\ell_A = 0$. Finally $w(A_L) = 18 + 6L + 3\binom{L}{2}$ and $w(N_L) = 12 + 12L$ give $\sum_t C_t = 2w(A_L) + 2w(N_L) - 2 = 2\Gamma_L$. □

Proof of Theorem 1.2. By Lemma 5.1 and Theorem 4.1 with $\bar{w} = 2\Gamma_L + 2$, Max-k-Cut is $\big(\frac{2\Gamma_L}{2\Gamma_L+2} - \delta, \frac{2\Gamma_L - 4}{2\Gamma_L+2} + \delta\big)$-hard, and $\rho = 1 - 2/\Gamma_L$. For $L = 0, 1, 2, 3, 4$, $\Gamma_L = 29, 47, 68, 92, 119$. □

Remark 5.2. We also confirmed Lemma 5.1 by computer, for $L \leq 6$ (Appendix B, which also lists the values for $L = 1$). Placing the universal vertices inside the gadgets is more economical than adding them to the final graph as in Section 6: combining Theorem 1.1 with Lemma 6.1 gives only 47/49 for Max-4-Cut and 35/36 for Max-5-Cut.

Remark 5.3 (How the family was found). Theorem 4.1 makes the design of a family a single linear program (Appendix B). Seeded with E and N augmented by universal vertices (the construction of Section 6 applied to the gadgets of Section 3), this LP returned the ratios 45/47 and 33/34; we then simplified the optimal solutions into the family above. Our column-generation runs for $k = 4, 5$ did not find better families.

6 Universal vertices and perfect completeness

The following reduction is due to Kann, Khanna, Lagergren and Panconesi [KKLP97, Sec. 3] (for $L = 1$) and Guruswami and Sinop [GS13, Sec. 2.3] (Appendix D of the arXiv version; for $L \leq 2$, where the base graph is K-colourable with $3 \mid K$ and the weights are d_v/K).

Lemma 6.1. *Let G be a weighted graph with total weight M and weighted degrees d_v, let $L \geq 1$ and $\kappa \geq 0$. Let H be obtained by adding vertices $z_1, \dots, z_L$, joining each z_i to every $v \in V(G)$*

with weight $d_v/3$, and joining every pair z_i, z_j with weight κ. For a graph G' let $\mathrm{unc}_k(G')$ be the minimum weight of monochromatic edges over k-colourings of G'. Then

$$\min\{\kappa,\ \mathrm{unc}_3(G)\} \ \le\ \mathrm{unc}_{3+L}(H) \ \le\ \mathrm{unc}_3(G),$$

and $w(H) = M(1 + \frac{2L}{3}) + \binom{L}{2}\kappa$. If G is 3-colourable then H is $(3+L)$-colourable.

Proof. The upper bound: give $z_1, \dots, z_L$ the colours $3, \dots, L+2$ and colour G optimally with $\{0,1,2\}$. For the lower bound, let χ be a $(3+L)$-colouring of H. If two z_i share a colour, χ loses at least κ. Otherwise the z_i use L distinct colours, and a vertex $v \in V(G)$ can share its colour with at most one z_i. If it does, the edge vz_i is monochromatic, of weight $d_v/3$. Among the three colours not used by the z_i, some colour is used by neighbours of v of total weight at most $d_v/3$; recolouring v with it does not increase the weight of monochromatic edges. Repeating this for all such v yields a colouring that uses only three colours on G, whose monochromatic weight is at least $\mathrm{unc}_3(G)$. The total weight follows from $\sum_v d_v = 2M$. □

If G comes from a (c,s)-hard instance of Max-3-Cut with $\Delta_c = 1-c$ and $\Delta_s = 1-s$, choose $\kappa = \Delta_s M$, so that $w(H) = m_L M$ with m_L as in (4). Then $\mathrm{unc}_{3+L}(H) \le \Delta_c M$ in the YES case and $\ge \Delta_s M$ in the NO case, so Max-$(3+L)$-Cut is hard within

$$\frac{m_L - \Delta_s}{m_L - \Delta_c}, \qquad m_L = 1 + \tfrac{2L}{3} + \tbinom{L}{2}\Delta_s. \tag{4}$$

Proof of Theorem 1.3. Apply (4) to the $(1, \frac{16}{17} + \varepsilon)$-hardness of 3-colouring of [AOTW14] ($\Delta_c = 0$, $\Delta_s = \frac{1}{17}$): the ratio is $1 - \frac{\Delta_s}{m_L} = 1 - \frac{3}{51+34L+3\binom{L}{2}}$, e.g. $\frac{82}{85}$ for $L=1$ and $\frac{119}{122}$ for $L=2$. Perfect completeness is preserved. □

Remark 6.2. Where do the previously stated constants come from? Guruswami and Sinop first prove a bound for K-colourable graphs, for every K divisible by 3, by a tensor-product construction, and then add $k - K \le 2$ universal vertices [GS13, Sec. 2.3]. Their bound for K colours loses a factor K in the gap compared with the bound for 3 colours that it is built from, and they use it also for $K = 3$, where this loss is unnecessary; [AOTW14] plugged their 16/17 into the resulting formula. (For $k = 3$ the formula gives only 50/51.) Starting from the 3-colour bound directly, (4) improves on the formula for every k from 4 to 20; even the 32/33 of [GS13] would give 54/55 for Max-4-Cut.

7 A barrier for gadgets from 3Lin(Z3) to Max-3-Cut

Optimality proofs for gadgets via LP duality go back to [TSSW00]; [HHM+17] proved an analogous optimality statement for gadgets from Chan's Hadamard predicates to $2\textsc{Lin}(\mathbb{Z}_2)$.

Theorem 7.1 (computer-assisted)**.** *For every family (I_0, I_1, I_2) of* Max-3-Cut *gadgets (with any number of auxiliary vertices, possibly using the global vertices),*

$$\max_{x \in \mathbb{Z}_3^3} S_{y^\circ}(x) \ \ge\ \tfrac{27}{29} \textstyle\sum_t C_t.$$

Hence Theorem 4.1 cannot give a ratio below 27/29 for Max-3-Cut*, and the family of Section 3 is optimal for it.*

The proof exhibits a single distribution μ^* on $\mathbb{Z}_3^3$ such that, for every type t and every type-t gadget, the average of $V_t(x, y^\circ)$ over $x \sim \mu^*$ is at least $\frac{27}{29} C_t$. Summing over t gives the theorem, since the maximum of S_{y° is at least its average.

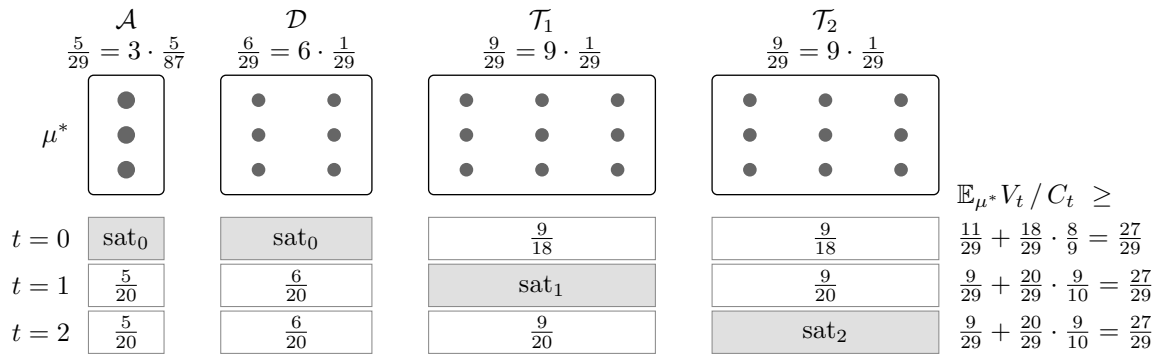


Figure 5: The distribution μ^* of the proof of Theorem 7.1. Top: one dot per tuple, at the positions of Fig. 3, with area proportional to its mass ($\frac{5}{87}$ on each tuple of $\mathcal{A}$, $\frac{1}{29}$ on every other tuple); the class masses are above the boxes. Row t: the shaded cells form sat_t, on which $V_t(x, y^\circ) \geq C_t$; each other cell gives the probability that x_r lies in that class, for $r \sim \nu$ from the type-t certificate of Lemma 7.2. These are exactly the conditional probabilities of the classes under μ^*, given that x is unsatisfying for type t. Right: the resulting lower bound on $\mathbb{E}_{\mu^*} V_t / C_t$, namely $\mu^*(\mathrm{sat}_t)$ plus the unsatisfying mass times the certificate's guarantee $\sum_x \theta(x)$ (Table 2); it is $\frac{27}{29}$ for every type.

The distribution μ^*. Partition $\mathbb{Z}_3^3$ into the classes $\mathcal{A}$ (all equal, 3 tuples), $\mathcal{D}$ (all distinct, 6 tuples), $\mathcal{T}_1$ and $\mathcal{T}_2$ (two equal with sum 1, resp. 2; 9 tuples each), as in Fig. 3, so that $\mathrm{sat}_0 = \mathcal{A} \cup \mathcal{D}$, $\mathrm{sat}_1 = \mathcal{T}_1$ and $\mathrm{sat}_2 = \mathcal{T}_2$. Let μ^* be uniform within each class, with class masses in the proportions

$$\mu^*(\mathcal{A}) : \mu^*(\mathcal{D}) : \mu^*(\mathcal{T}_1) : \mu^*(\mathcal{T}_2) = 5 : 6 : 9 : 9,$$

that is, mass $\frac{5}{29}$ on $\mathcal{A}$ and $\frac{1}{29}$ on every other tuple (Fig. 5). A type-t gadget is guaranteed its completeness value C_t on sat_t. On the unsatisfying tuples we use a certificate (Lemma 7.2) that guarantees on average a fraction $\frac{8}{9}$ of C_0 for type 0 and $\frac{9}{10}$ of C_t for $t = 1, 2$ (Table 2), but only for one particular distribution of inputs: uniform on $\mathcal{T}_1 \cup \mathcal{T}_2$ for type 0, and in the proportions $5 : 6 : 9$ on $\mathcal{A}, \mathcal{D}, \mathcal{T}_2$ for type 1 (on $\mathcal{A}, \mathcal{D}, \mathcal{T}_1$ for type 2). These requirements determine μ^*: it is the only distribution, uniform within each class, whose conditional distribution on the unsatisfying tuples of each type is that of the certificate. With these masses all three types end at the same lower bound on $\mathbb{E}_{\mu^*} V_t / C_t$, namely $\mu^*(\mathrm{sat}_t)$ plus the unsatisfying mass times the certificate's fraction: $\frac{11}{29} + \frac{18}{29} \cdot \frac{8}{9} = \frac{9}{29} + \frac{20}{29} \cdot \frac{9}{10} = \frac{27}{29}$.

Certificates. The bounds on the unsatisfying tuples are instances of weak LP duality.

Lemma 7.2. *Fix t and let $\mathcal{F}_t = [3]^{\mathrm{sat}_t}$ be the set of* columns *(functions from sat_t to colours). Suppose $\theta \colon \mathrm{sat}_t \to \mathbb{R}_{\geq 0}$ and a probability distribution ν over pairs $r = (x_r, \chi_r)$, with $x_r \in \mathbb{Z}_3^3$ and χ_r a colouring of the* universe $\Omega = \{p_1, p_2, p_3, g_0, g_1, g_2\} \cup \mathcal{F}_t$ *with $\chi_r(p_j) = (x_r)_j$ and $\chi_r(g_i) = i$, satisfy, for every pair $\{\xi, \eta\} \subseteq \Omega$,*

$$\Pr_{r \sim \nu}[\chi_r(\xi) \neq \chi_r(\eta)] \;\geq\; \sum_{x \in \mathrm{sat}_t} \theta(x)\, [\mathrm{can}_x(\xi) \neq \mathrm{can}_x(\eta)], \tag{5}$$

where can_x is the canonical colouring of Ω at x: $\mathrm{can}_x(p_j) = x_j$, $\mathrm{can}_x(g_i) = i$, $\mathrm{can}_x(f) = f(x)$. Then every type-t gadget satisfies $\mathbb{E}_{r \sim \nu} V_t(x_r, y^\circ) \geq \big(\sum_x \theta(x)\big)\, C_t$.

Proof. For each $x \in \mathrm{sat}_t$ fix a colouring of the auxiliary vertices of I_t attaining $V_t(x, y^\circ) \geq C_t$. Let $\phi \colon V(I_t) \to \Omega$ map p_j and g_i to themselves and each auxiliary vertex to its column (its colour

type t	sat_t	distribution of x_r (by class)	$\sum_x \theta(x)$	rows	common denominator
0	$\mathcal{A} \cup \mathcal{D}$	uniform on $\mathcal{T}_1 \cup \mathcal{T}_2$	8/9	78	324
1	$\mathcal{T}_1$	$\mathcal{A}, \mathcal{D}, \mathcal{T}_2$ in proportions $5:6:9$	9/10	162	4680
2	$\mathcal{T}_2$	$\mathcal{A}, \mathcal{D}, \mathcal{T}_1$ in proportions $5:6:9$	9/10	162	4680

Table 2: The certificates of Lemma 7.2 used for Theorem 7.1. In each, x_r is uniform within each class and never satisfying; "rows" is the number of pairs r (the *rows* of the certificate) with positive weight under ν, and the last column is the common denominator of the stored exact certificate (Appendix B).

as a function of x). Then $\mathrm{can}_x \circ \phi$ is the fixed optimal colouring at x, and $\chi_r \circ \phi$ is a colouring with primaries x_r and globals y°, of value at most $V_t(x_r, y^\circ)$. Summing (5) over the edges $\xi'\eta'$ of I_t with $\phi(\xi') \neq \phi(\eta')$, with their non-negative weights, gives $\mathbb{E}_r V_t(x_r, y^\circ) \geq \sum_x \theta(x) V_t(x, y^\circ) \geq (\sum_x \theta(x))\, C_t$; edges with $\phi(\xi') = \phi(\eta')$ are never cut by $\mathrm{can}_x \circ \phi$. □

Proof of Theorem 7.1. We use certificates as in Lemma 7.2 with the parameters of Table 2. The type-2 certificate is obtained from the type-1 certificate by symmetry: for a type-2 gadget I, let I' be I with g_1 and g_2 exchanged. By invariance under the colour permutation $c \mapsto -c$, $V_{I'}(x, y^\circ) = V_I(-x, y^\circ)$, so I' is a type-1 gadget with the same completeness, and the type-1 certificate gives $\mathbb{E}\, V_I(-x_r, y^\circ) \geq \frac{9}{10} C_I$, where $-x_r$ is distributed as in the last row of Table 2. (We also stored and checked this mirrored certificate explicitly.)

In each certificate, the distribution of x_r is the conditional distribution of μ^* on the complement of sat_t. Hence, since $V_t(x, y^\circ) \geq C_t$ on sat_t,

$$\mathbb{E}_{\mu^*} V_0 \geq \tfrac{11}{29} C_0 + \tfrac{18}{29} \cdot \tfrac{8}{9} C_0 = \tfrac{27}{29} C_0, \qquad \mathbb{E}_{\mu^*} V_1 \geq \tfrac{9}{29} C_1 + \tfrac{20}{29} \cdot \tfrac{9}{10} C_1 = \tfrac{27}{29} C_1,$$

and likewise for V_2. Summing over t, $\max_x S_{y^\circ}(x) \geq \mathbb{E}_{\mu^*} S_{y^\circ}(x) \geq \frac{27}{29} \sum_t C_t$. □

Remark 7.3. In each row r of the certificates, the colour of a column f is $h_r(f|_{R_r})$ for a set R_r of one or two satisfying tuples and a table $h_r \colon [3]^{R_r} \to [3]$. The certificates were found by solving the dual LP with constraint generation over pairs, and then rounded to rationals (Table 2). The rounded certificates are what is verified: (5) was checked *exactly*, for all pairs of elements of Ω ($|\mathcal{F}_t| = 3^9$), by two programs using different methods; the minimum slack is 0 (Appendix B).

8 Discussion

- *Perfect completeness.* Our reductions lose completeness (29/30); on 3-colourable graphs 16/17 [AOTW14] is still the best bound (the first constant bound there is due to [Pet94]). Can the negation trick be combined with a source that has perfect completeness?
- *Other sources.* By Theorem 7.1 and Proposition 4.3, per-type gadgets on the instances of Theorem 2.1, analysed through Theorem 2.1 alone, cannot beat 27/29. Natural candidates for other sources are the predicates $\sum_i a_i x_i = t$ over $\mathbb{Z}_3$ with $a \in \{\pm 1\}^r$ and $\sum_i a_i \equiv 0$, which are NP-hard to approximate by Chan's theorem [Cha16] (the unconditional analogue of [AM09]) and for which the negation trick again yields colour-symmetric predicates; gadget reductions from Chan's predicates have been used successfully for $2\textsc{Lin}(\mathbb{Z}_2)$ [HHM$^+$17]. We expect the proof of Theorem 4.1 to carry over to Chan's r-partite instances, with one decoding map per player (the analogue of Lemma 4.2 for products of r arbitrary vectors), in which case no symmetry hypothesis would be needed.

- *Optimality for $k \geq 4$.* We do not know whether 45/47 and 33/34 are optimal under Theorem 4.1; our column generation found nothing better, but we have no barrier certificate.
- *Large k.* Our family gives $1 - \Theta(1/k^2)$; combining Theorem 4.1 with the tensor construction of [GS13] might improve the constant in $1-\Theta(1/k)$ and the range $k \leq 20$ in which we improve the known bounds.

A Further remarks on the soundness criterion

Remark A.1 (The symmetry hypothesis). The symmetry hypothesis of Theorem 4.1 is harmless: replace I_t by $I'_t = \frac{1}{2}I_t \sqcup \frac{1}{2}I_t^{\mathrm{sw}}$, where I_t^{sw} is I_t with p_2, p_3 exchanged and $\sqcup$ takes disjoint copies of the auxiliary vertices while sharing primaries and globals. Then $V'_t(x,y) = \frac{1}{2}(V_t(x,y)+V_t(x_1,x_3,x_2,y))$, so no C_t decreases (the satisfying sets are symmetric) and $\max_{x,y} S_y(x)$ does not increase. Gadgets that are symmetric under all permutations of the primaries, as all gadgets in this paper, satisfy the hypothesis trivially.

Remark A.2 (Balance). The proof of Theorem 4.1 does not use that the types are balanced in the NO case; with constant maps $a \equiv c_1$, $b \equiv c_2$ the soundness hypothesis in fact forces every type to have weight $\frac{1}{3} \pm O(\varepsilon)$.

Remark A.3 (Comparison with the per-type analysis). For $k = 3$, Theorem 4.1 is never worse than the per-type analysis. Every $x \in \mathbb{Z}_3^3$ satisfies exactly one type $t(x)$, so $S_y(x) \leq V^+_{t(x)} + \sum_{t \neq t(x)} V_t^-$, where V_t^+ and V_t^- are the maxima of V_t over satisfying resp. unsatisfying x and all y; hence $\rho \leq (\sum_t V_t^- + \max_t(V_t^+ - V_t^-))/\sum_t C_t$, which is the bound of the per-type reduction theorem [NRT25, Thm. C.2]. With $V_t^+ = C_t$ and $V_t^- = C_t - d$ this is $1 - 2d/\sum_t C_t$. Theorem 4.1 can be much better because a tuple that is unsatisfying for two types need not be bad for both gadgets simultaneously.

Proposition A.4. *For every family for* Max-k-Cut*, $\rho \geq \alpha_k$, where α_k is the approximation ratio of the Frieze–Jerrum rounding [FJ97, dKPW04].*

Proof. Let $\zeta_0, \ldots, \zeta_{k-1}$ be unit vectors with $\langle \zeta_c, \zeta_{c'} \rangle = -\frac{1}{k-1}$ for $c \neq c'$. For each t and each satisfying tuple x of type t (there are 9), fix a colouring χ_x^t of I_t with primaries x, globals y° and value $V_t(x, y^\circ) \geq C_t$, and let $v_\xi^t = (9^{-1/2}\zeta_{\chi_x^t(\xi)})_x$ for each vertex ξ of I_t. These are unit vectors with $\langle v_\xi^t, v_{\xi'}^t \rangle = 1 - \frac{k}{k-1}\Pr_x[\chi_x^t(\xi) \neq \chi_x^t(\xi')] \geq -\frac{1}{k-1}$, and their Frieze–Jerrum objective $\sum w_{\xi\xi'} \frac{k-1}{k}(1 - \langle v_\xi^t, v_{\xi'}^t \rangle)$ equals $\mathbb{E}_x V_t(x, y^\circ) \geq C_t$. On the set B of primary and global vertices, the disagreement probabilities do not depend on t (any two primaries are independent and uniform on $\{0,1,2\}$), so the Gram matrix of $(v_\xi^t)_{\xi \in B}$ is the same for all t. Fix vectors $(f_\xi)_{\xi \in B}$ in some $\mathbb{R}^m$ with this Gram matrix, let Ψ_t be the isometry from $\mathrm{span}\{v_\xi^t : \xi \in B\}$ onto $\mathrm{span}\{f_\xi\}$ with $v_\xi^t \mapsto f_\xi$, and Π_t the orthogonal projection onto the former span. In a direct sum $\mathbb{R}^m \oplus H_0 \oplus H_1 \oplus H_2$ put $\tilde{v}_\xi = \Psi_t \Pi_t v_\xi^t + \iota_t(v_\xi^t - \Pi_t v_\xi^t)$ for ξ in I_t, where ι_t embeds into H_t. Then $\tilde{v}_\xi = f_\xi$ on B and all inner products within each I_t are preserved, so this is a vector solution for $I_0 \cup I_1 \cup I_2$ (glued along B) with objective at least $\sum_t C_t$, in which every edge has inner product $\geq -\frac{1}{k-1}$. (Auxiliary vertices of different gadgets are not adjacent, so their inner products do not matter.) The rounding of [FJ97, dKPW04] cuts each such edge with probability at least $\alpha_k \frac{k-1}{k}(1 - \langle \tilde{v}_\xi, \tilde{v}_{\xi'} \rangle)$, so some colouring has value at least $\alpha_k \sum_t C_t$; but every colouring has value at most $\max_{x,y} S_y(x)$. □

B Methodology and computations

How the results were obtained. The research was carried out with an AI system (Claude, by Anthropic) working in an agentic coding environment, under the direction of the author, who set

the goals at each stage (first improving the gadgets of [NRT25] for Max-4-Cut and Max-5-Cut, then attacking Max-3-Cut) and decided which directions to pursue. The AI system wrote and ran the linear-programming searches for gadgets and the verification programs, proposed the negation trick of Section 3 while exploring alternative source problems, derived the soundness criterion of Section 4, found the gadget families of Section 5 (by the linear program described below, followed by simplification) and the hand proof of Lemma 5.1, observed the consequences of the universal-vertex reduction in Section 6, computed the certificates of Section 7, searched the literature, and drafted the text. Every main claim was checked in two ways. First, independently instantiated AI agents, which did not see the reasoning that produced a claim, were asked to refute it: they re-read the primary sources (in particular Håstad's verifier and the constants of [GS13, AOTW14]), re-derived the proofs, and re-implemented the computations from scratch; four such referees also reviewed a draft of this paper. Second, every gadget value and every certificate was verified by at least two independently written programs, all using exact arithmetic, as described below.

Availability. The code, the gadget families and the certificates are available at `https://github.com/ziangni-sys/max-3-cut`. All programs are in Python 3 with NumPy; HiGHS was used for the linear programs.

Verification of families. `scripts/verify_sum_criterion.py`, applied to the families produced by `scripts/make_unified_family.py`, computes $V_t(x, y)$ for all $x \in [k]^3$ and $y \in \mathcal{Y}_k$ by exhaustive enumeration of the auxiliary colourings in integer arithmetic, checks the symmetry hypothesis of Theorem 4.1, and outputs the exact ratio ρ; it reproduces Lemma 5.1 for $L \le 3$. The independently written `scripts/independent/family_dp.py` (which optimises the auxiliary vertices of A_L by dynamic programming along the 6-cycle) reproduces it for $L \le 6$. For $L = 1$, Table 3 lists the values by the pattern of equalities among x_1, x_2, x_3 and $\mathrm{col}(z_1)$; the remaining patterns follow by permuting the primaries and the colours.

$(x_1, x_2, x_3 \mid \mathrm{col}(z_1))$	$(0,0,0 \mid 0)$	$(0,0,0 \mid 1)$	$(0,0,1 \mid 0)$	$(0,0,1 \mid 1)$	$(0,0,1 \mid 2)$	$(0,1,2 \mid 0)$	$(0,1,2 \mid 3)$
V_{2A_1}	48	48	48	48	44	48	48
V_{N_1}	12	21	17	20	23	21	21
$S = V_{2A_1} + 2V_{N_1}$	72	90	82	88	90	90	90

Table 3: Gadget values of the family $(2A_1, N_1, N_1)$ for Max-4-Cut ($L = 1$), by colour pattern.

`scripts/verify_sym3.py` checks Lemma 3.1 and the identity (3); the end-to-end reduction of Section 3 was also tested on random small instances of 3Lin($\mathbb{Z}_3$) by brute force.

The design LP. For fixed candidate auxiliary vertices, each described by its column (its colour in the completeness colouring at each satisfying tuple), each type-t block has one variable per orbit of edges under permutations of the primaries. Completeness rows require the canonical colouring at each satisfying tuple to have value at least c_t; soundness is enforced by variables $v_t(x, y)$ with rows $v_t(x, y) \ge$ (value of an explicit colouring), generated by local search and verified by exact separation, and coupling rows $\tau \ge \sum_t v_t(x, y)$ for all (x, y); we minimise τ subject to $\sum_t c_t \ge 1$. New auxiliary vertices were generated by exact dual pricing of single vertices (over all 4^9 columns for $k = 4$) and by pricing whole small gadgets with up to 7 auxiliary vertices; in our runs the optimum stayed at 45/47 ($k = 4$) and 33/34 ($k = 5$).

Barrier certificates. The certificates of Theorem 7.1 are stored as exact integers (common denominator D, row weights, θ, and for each row x_r, R_r, h_r) in the files `cert_t0.json`, `cert_t1.json` and `cert_t2.json` of the directory `results/barrier_k3`. The checker `check_exact.py` verifies that $\theta \geq 0$, that ν is a probability distribution whose marginal on x_r is uniform within each class, that $\sum \theta$ equals $\frac{8}{9}$ resp. $\frac{9}{10}$, and verifies (5) for all pairs of elements of Ω, including all $\binom{3^9}{2}$ pairs of columns, with integer-valued floating-point matrix products whose partial sums stay below 2^{53} (hence exact). A second, independently written checker (`check_exact_pairwise.py`) writes the slack of a pair of columns (f, f') as a sum of unary and pairwise terms in the nine pairs $(f(x), f'(x))$, $x \in \mathrm{sat}_t$, and minimises it exactly over all 9^9 choices of these pairs. Both find minimum slack 0.

Universal vertices. `scripts/universal_vertex.py` checks Lemma 6.1 by brute force on random weighted graphs and computes (4).

AI methodology

We used Claude (Anthropic), a large language model, in an agentic coding environment (Claude Code, including multi-agent workflows), to carry out this research and to write this paper. The tool proposed the main ideas (the negation trick, the soundness criterion and its proof, the gadget families and the proof of Lemma 5.1, the use of the universal-vertex reduction, and the barrier certificates), wrote and ran all search and verification programs, performed the literature search, and drafted the text of all sections. It therefore materially affected all sections of the paper; the methodology, including how the claims were checked by independent AI agents and by independent programs, is described in Appendix B. The author directed the research and is responsible for the content of the submission. The author verified the correctness and originality of all content, including references.

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